

M003 Week 3 — English Worksheet (Beginner)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you draw **pictures with blocks** on the wall. A picture is made of **rows**. A loop inside a loop draws the picture row by row: the inner loop grows **x** across, then the outer loop grows **y** up to the next row.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
set y to 0
repeat 2 times:
  set x to 0
  repeat 3 times:
    place red block at (x, y)
    add 1 to x
  add 1 to y
```

Does this picture have 2 rows, one above the other? Circle one: yes · no

Why?


```
set y to 0
repeat 2 times:
  set x to 0
  repeat 3 times:
    place red block at (x, y)
    add 1 to x
```

This time y never grows. Do the two rows land on top of each other? Circle one: yes · no

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — After each row, the picture should step **up** to the next y.

```
# clean
set y to 0
repeat 3 times:
  set x to 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to x
  add 1 to y
```

```
# buggy
set y to 0
repeat 3 times:
  set x to 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to x
```

What is wrong? Where do all the rows land?

Pair B — The bottom row should look like **red, red, blue**.

```
# clean
place red block at (0, 0)
place red block at (1, 0)
place blue block at (2, 0)
```

```
# buggy
place blue block at (0, 0)
place red block at (1, 0)
place red block at (2, 0)
```

What is wrong?

3 · Fill the Gap

The inner loop should make each row grow **across**. One word is missing. Fill it in using the word bank.

```
set y to 0
repeat 3 times:
  set x to 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to ____
  add 1 to y
```

Word bank: x · y · block

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. Draw a small **pixel art picture** on the wall — a heart, a flower, or a letter — using rows of colored blocks. When you finish, come back here.

Send a photo or video of your picture, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My picture is a ...

The colors in the bottom row are ...

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your picture on the wall. Talk like you are teaching a friend. Try to use these words: **row**, **loop**, **pattern**, **color**.

1. Show your finished picture.
2. Point at the bottom row and say the colors in order.
3. Say in your own words how the rows stack up to make a picture.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.